MIL-P-15736/1D(SH)
21 September 1987
SUPERSEDING
MIL-P-15736/1C(SHIPS)
14 May 1973
(See 6.3)

MILITARY SPECIFICATION

POWER SUPPLY, METALLIC RECTIFIER, HELICOPTER STARTING AND SERVICING

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers equipments for converting power from 440 volts, 60 hertz (Hz), 3-phase, alternating current (ac) to 28 volts direct current (dc). The dc output power is regulated and controlled as required for helicopter servicing, reciprocating and gas turbine engine starting.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose).

MILITARY

MIL-R-22 - Resistors, Variable (Wirewound, Power Type), General Specification for.
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-C-2212 - Controllers, Electric Motor A.C. or D.C., and Associated Switching Devices.
MIL-L-3661 - Lampholder, Indicator Lights, Indicator-Light Housings, and Indicator-Light Lenses, General Specification for.
MIL-L-3661/54 - Lenses, Indicator Light, Style LC40.
MIL-L-3661/60 - Lampholder, Transformer, Light, Indicator, Style LH93.
MIL-L-3661/61 - Lampholder, Lights, Indicator (Housing), Style LH94.
MIL-L-3661/64 - Lampholder, Lights, Indicator (Housing), Style LH97.
MIL-L-3661/65 - Lampholder, Lights, Indicator (Housing), Style LH98.
MIL-S-3950 - Switches, Toggle, Environmentally Sealed, General Specification for.
MIL-M-10304 - Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for.
MIL-R-15109 - Resistors and Rheostats, Naval Shipboard.
MIL-L-15719 - Lubricating Grease (High-Temperature, Electric Motor, Ball and Roller Bearings).
MIL-P-15736 - Power Supplies, Metallic Rectifier, Naval Shipboard.
MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-701 - Lists of Standard Semiconductor Devices.
MIL-STD-1399, Section 103 - Interface Standard For Shipboard Systems Electric Power, Alternating Current.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General. The power supply shall be in accordance with class R of MIL-P-15736 and as specified herein.

3.2 Electrical ratings and performance. Electrical ratings and performances shall be as specified in table I and 3.2.1 through 3.2.5.

TABLE I. Electrical ratings.

Parameter	Requirement
Ac volts (input)	440 ± 5 percent
Phase (input)	3
Dc volts (output)	28 (nominal - see 3.2.1)
Dc amperes (output)	See 3.2.2
Noise and ripple voltage content, maximum	2 volt peak to peak (see 3.2.5)
Regulation	2 percent (see 3.2.2)
Type of load	Resistive, inductive (including motors) and capacitive

3.2.1 Output voltage adjustment and control. Dc output voltage shall be continuously adjustable from 26 to 32 volts, under any specified condition of input voltage and frequency and at any value of output current up to 450 amperes. The normal installation of power supplies covered by this specification is shown on figure 1. The resistance values are maximum values and include resistance of both positive and negative conductors. The power supply may use an adjustable compounding circuit which yields a rise in output terminal voltage as current increases, or a combination of such compounding control and remote sensing. However, remote sensing leads shall not be connected to a point closer to the load than the junction box shown on figure 1. The objective of the compounding circuit specified herein and its associated adjustment device (see 3.8.4(d)) is to provide compensation for voltage drop in load cables consequent to load current.

3.2.2 Servicing loads. The power supply shall have a continuous duty rating of at least 300 amperes. Output voltage shall be automatically regulated to within plus or minus 2 percent of the no load value within the range of 26 to 32 volts from no load to 450 amperes.

3.2.3 Reciprocating engine starting. The power supply shall furnish the starter motor current requirements shown on figure 2. The value to which the voltage drops upon application of the engine starting load is not critical, but the voltage measured at the load shall recover to and remain above 20 volts (but shall not exceed the preset value) within 200 milliseconds (0.2 second) of initial application. Two consecutive such cycles shall occur with a no load rest period of 30 seconds duration between cycles.

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3.2.4 Gas turbine engine starting. The power supply shall furnish the starter motor current requirements shown on figure 3. The value to which the voltage drops upon application of the engine starting load is not critical, but the voltage measured at the load shall recover to, and remain above, 20 volts (but shall not exceed the preset value) within 200 milliseconds (0.2 second) of initial application. A minimum of four consecutive starts shall be required. A 5-minute cooling period shall elapse following four such starts, prior to any subsequent loading of the power supply.

3.2.5 Ripple and noise. The combined noise and ripple present in the power supply output voltage shall not exceed 2.0 volts peak-to-peak in magnitude at 28 volts dc at point of regulation at no load, 150 amperes load, and 300 amperes load. This performance shall be satisfied under conditions of balanced line input.

3.2.6 Ripple. Ripple content of power supply output voltage shall not exceed 1/2 volt root mean square (rms) at 28 volts dc at point of regulation, from no load to 300 amperes load. This requirement shall apply for all conditions of steady state line voltage unbalance specified for type I system characteristics in accordance with MIL-STD-1399, section 103.

3.3 Duty. Duty shall be continuous and intermittent (see 3.2.2, 3.2.3 and 3.2.4).

3.4 Ambient temperature. The operating ambient temperature range shall be 20 to 50 degrees Celsius (°C).

3.5 Ventilation. Ventilation shall be forced draft (fan) or natural draft.

3.6 Mounting. The equipment shall be constructed for deck mounting.

3.7 Meters. Meters shall be as follows:

(a) Dc voltmeter (0-50 volts).
(b) Dc ammeter (0-500 amperes).

3.8 Parts.

3.8.1 Semiconductors. Semiconductor devices shall be silicon types and shall be selected from MIL-STD-701. These devices shall comply with the requirements of MIL-S-19500.

3.8.2 Terminal boards. Terminal boards shall be provided for input, output and control connections.

3.8.3 Magnetic contactor. A magnetic contactor complete with pushbutton station, complying with MIL-C-2212 shall be furnished and connected to provide on-off control of the power supply.

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3.8.4 Miscellaneous parts within power supply. Miscellaneous parts as follows shall be installed within the power supply.

(a) A rheostat, conforming to MIL-R-22 or MIL-R-15109, connected to permit continuous adjustment of the output voltage over the specified range (see 3.2.1).
(b) A 500-ampere 50-millivolt shunt, to permit the ammeter of 3.8.5(c) to measure output current.
(c) Terminal blocks to permit connection of remote "start" and "stop" pushbuttons (see 3.8.5(a)).
(d) A rheostat, similar to that of (a) herein, as necessary to adjust the voltage compounding control required by 3.2.1 over the range of cable resistance from zero to the resistance values shown on figure 1. The purpose of this control device and associated circuits is to provide load voltage compensation for voltage drop in cables in which voltage drop compensation cannot be provided by remote sensing.

3.8.5 Miscellaneous parts outside power supply. Miscellaneous parts as follows shall be installed on the power supply enclosure exterior:

(a) "Start" and "stop" pushbuttons (see 3.8.3).
(b) An ammeter "press-to-read" switch, spring return to normal position, conforming to MIL-S-3950 arranged to connect the ammeter specified below to the shunt, when depressed.
(c) An ammeter and a voltmeter 3-1/2 inch round panel type conforming to MIL-M-10304. Meter ranges shall be as specified in 3.7.
(d) An indicator light, conforming to MIL-L-3661 and style LH93 of MIL-L-3661/60, LH94 of MIL-L-3661/61, LH97 of MIL-L-3661/64 or LH98 of MIL-L-3661/65 with white lens in accordance with style LC40 WT3 of MIL-L-3661/54, connected to indicate "Power on".

3.9 Size and weight. Size and weight of power supplies shall not exceed the following values.

Width	24 inches.
Height	48 inches.
Depth	24 inches.
Weight	850 pounds.

3.10 Forced draft ventilation (if used, see 3.5). The following shall apply:

(a) Fan and fan motor may conform to commercial practice, except that electrical insulation and corrosion resistant treatment shall comply with MIL-E-917.
(b) Fan and motor bearings shall be sealed types requiring no lubrication. Preferred types of bearings are type 120 or type 111, class 3 or class 8 of FF-B-171. Preferred lubricant is silicone grease conforming to MIL-L-15719.

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(c) Screens and warning plates shall be used to minimize the probability and severity of personal injury due to contact with moving fan blades.
(d) Fan and motor shall sustain no damage during shock and vibration tests of the assembled equipment.

3.11 Airborne noise. Airborne noise requirements of MIL-P-15736 shall apply except that the limit applicable to power supplies specified herein shall be 10 decibels (dB) (ref. 10^{-12} watts) greater than the limits specified in MIL-P-15736.

3.12 Enclosure. The enclosure shall be dripproof protected with a hinged front access door.

3.13 Manuals. Manuals shall be prepared as specified in MIL-P-15736. The manual shall contain oscillograms of the power supply output voltage showing normal ripple and noise content. The oscillograms shall be labeled with vertical and horizontal amplitudes to facilitate reproducing such observations with an oscilloscope.

3.14 Safety. The power supply shall be constructed in accordance with the safety requirements of MIL-E-917 in order to ensure maximum personnel safety and minimum casualties to equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Sampling, examination and tests. Sampling, examination and tests, to determine conformance with this specification, shall be conducted as specified in MIL-P-15736 and as specified herein.

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4.3 Qualification and periodic tests. Duty cycles specified in 3.2.3 and 3.2.4 shall be repeated four times to establish that the power supply parts do not exceed their safe temperature ratings.

4.3.1 Transient voltage regulation. The gas turbine engine starting cycle shown on figure 3 shall be simulated as specified herein at rated and low (see table I) line voltage. Cable resistances shall be as shown on figure 1. Remote regulation of dc output voltage shall be obtained by means of remote sensing or output voltage compensation. If remote sensing is employed for output voltage regulation, remote sensing leads extending from the power supply shall be connected at the junction box shown on figure 1. Preceding the load transient regulation test defined herein, a no load set point of 28 volts dc shall be established at the load terminals shown on figure 1. Voltage compensation control, if used, shall be adjusted for optimum performance as specified in 3.2.4. Initial conditions shall be recorded. Without any further adjustment, the system simulated in accordance with figure 1 shall be restored to no load operation, with the specified cable connected to the power supply. A 900-ampere resistive load shall be applied in a single step at the end of the load cable. The 900-ampere load shall be maintained for 2 seconds. At the end of this 2-second interval, the 900-ampere load shall be reduced in a single step to 450 amperes; this 450-ampere load shall be sustained for 28 seconds. The 450-ampere load shall then be removed in a single step, and the power supply shall be restored to no load operation. The minimum value to which load voltage decays, upon step increase of load, shall be recorded; load voltage shall also be recorded throughout the 2-second interval during which 900-ampere load is impressed, and the 28-second interval during which 450-ampere load is applied. Any overshoot in load voltage which is produced by each specified step decrease of load shall be reported. Load voltage waveforms associated with step change of load shall be recorded oscillographically in order to display the transient response characteristics of the power supply. Output current shall also be recorded by means of an oscillograph to display step application and step removal of load. This test shall demonstrate conformance to 3.2.4 and shall also verify conformance to 3.2.3.

4.3.2 Ripple and noise. The power supply shall be subjected to an observation of its output voltage with an oscilloscope at current values of zero, 150 and 300 amperes. Oscilloscope sweep rates, CRT persistence and an oscilloscope-mounted camera with high-speed film shall be adjusted to detect noise and ripple content of the ouput voltage of frequencies up to 1 megahertz. Noise and ripple waveform shall be photographically recorded. In the event that silicon controlled rectifiers are used for control and regulation of the power supply output voltage, a particular effort shall be made to detect spikes due to switching of these devices. Presence of noise or ripple exceeding 2 volts peak-to-peak in magnitude shall be cause for rejection.

4.4 Selective, periodic and qualification tests.

4.4.1 Ripple (rms). Ripple content shall be determined under each condition of unbalanced input line voltage specified in MIL-P-15736. Effective rms value of ripple voltage shall be measured by means of a vacuum tube voltmeter, calibrated in rms values, in series with a blocking capacitor. This examination shall verify 3.2.6.

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4.5 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.2.)

5.1 Preservation-packaging, packing and marking. Preservation-packaging, packing, and marking for shipment and storage shall be as specified in MIL-P-15736.

6. NOTES

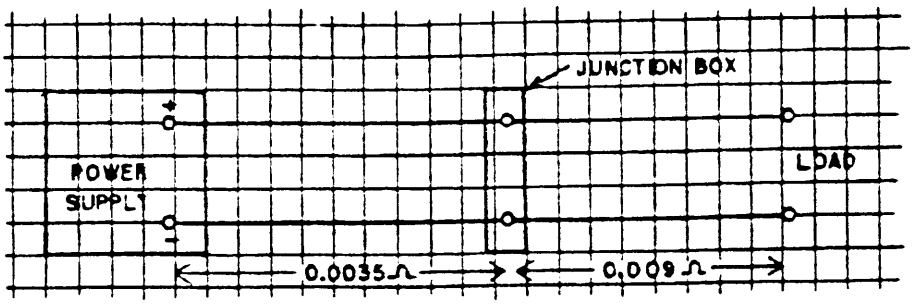
6.1 The notes specified in MIL-P-15736 are applicable to this specification.

6.2 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

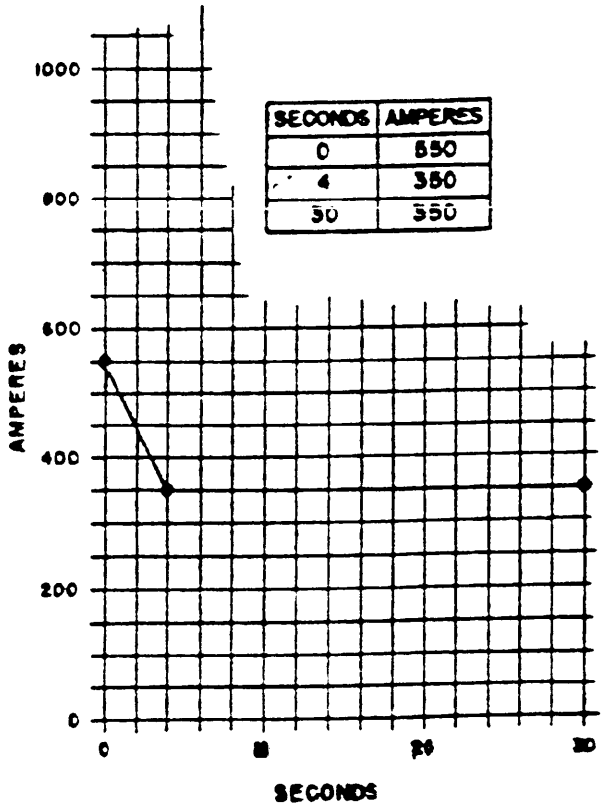
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SH 9651

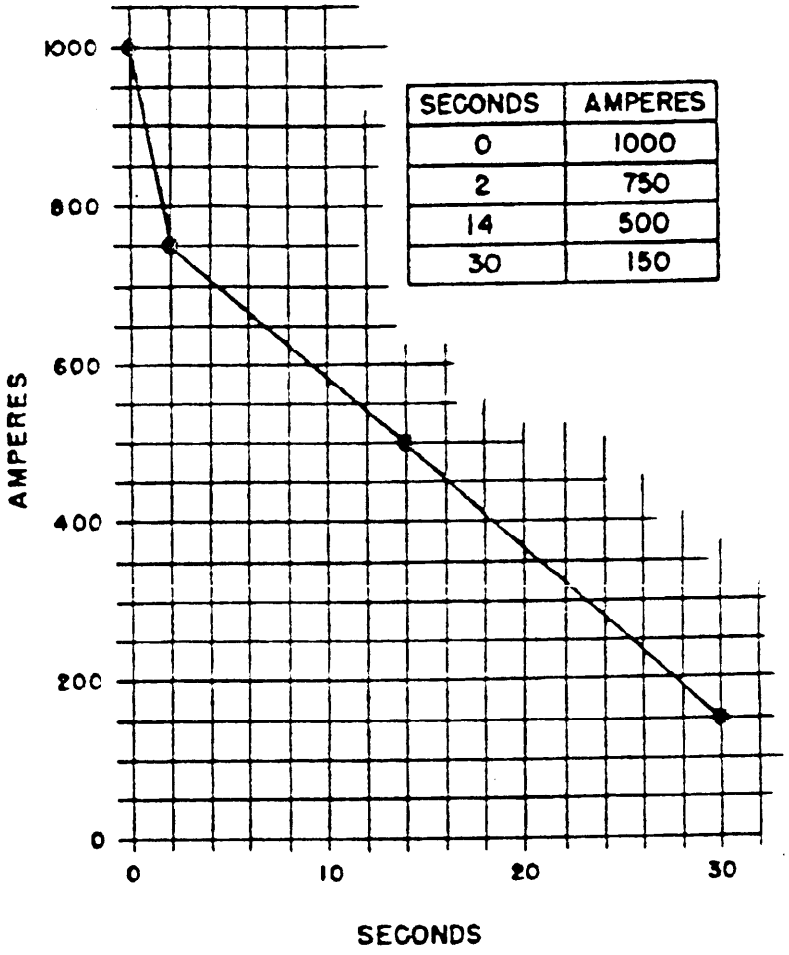
FIGURE 1. Cable resistances (ohms).



SH 9652

FIGURE 2. Reciprocating engine starter motor requirements.

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SH 9653

FIGURE 3. Gas turbine engine starter motor requirements.

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(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-15736/1D(SH)	POWER SUPPLY, METALLIC RECTIFIER, HELICOPTER STARTING AND SERVICING

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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